Uganda Institute of Information and Communications Technology

"In Quest of Excellence"

BSM121: RESEARCH METHODS

MODULE 1: Introduction to Research

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Definition of Research

- The word “research” originated from the old French word “researcher” meaning to search and search again. Research in common parlance refers to a **search for knowledge.**

- Research may be broadly defined “as a systematic inquiry towards understanding a complex social phenomenon or a process”.

- **OR** “Research is a systematic investigation to describe, explain, predict, and control the observed phenomenon related to a particular problem/area”.

- An approach becomes systematic when a researcher follows the scientific method. Research is systematic because it follows certain steps that are logical in order.

- Research follows **a scientific method,** which is an objective, logical, and systematic method, i.e., a method free from personal bias or prejudice

Role of Research

1. **Advance knowledge:** Research shares **new findings**, theories, and innovations with the global academic and professional community. It helps expand the **scientific and intellectual base** in a particular field.
2. **Problem-solving**: It provides solutions to specific challenges in fields like healthcare, agriculture, technology, and more. For instance, research can lead to the development of vaccines to combat diseases.
3. **Informed decision-making**: Research equips policymakers, organizations, and individuals with data and insights to make evidence-based decisions. For example, research on climate change influences environmental policies.
4. **Innovation and development**: It drives technological advancements and product development. Research in renewable energy, for instance, has led to more efficient solar panels and wind turbines.

Role of Research

5. **Testing hypotheses**: Research validates or refutes existing theories and beliefs, ensuring accuracy and scientific rigor. For example, it helps confirm the effectiveness of a drug through clinical trials.
6. **Understanding trends**: Research identifies patterns and trends in society, markets, or industries, which can guide future planning. For example, market research helps businesses anticipate consumer needs.
7. **Teaching and Learning:** Research supplies educators and students with reliable material. It fuels curricula, guides mentorship, and instills critical thinking, problem solving—an intellectual inheritance passed across generations
8. **Social and cultural development**: Research can address social issues, promote equality, and preserve cultural heritage. For example, studies in anthropology uncover the traditions and values of ancient civilizations.

Qualities of a Good Research

All types of research follow the scientific method which satisfies the following criteria:

- **Problem-Oriented**: It is driven by the desire to address specific questions, gaps, or challenges, leading to solutions or advancements.
- **Systematic:** It means that research is structured with specified steps to be taken in a specified sequence in accordance with the well-defined set of rules.
- **Logical**: Research is rooted in logical reasoning, using sound principles to draw conclusions from data and evidence.
- **Uses relevant, empirical data and proper data analysis methods:** Empirical data is data that has been collected by researchers themselves through observation, experience, or experimentation (Bradford, 2017).
- **Objective**: Research aims to uncover facts and truths without bias, relying on evidence-based analysis rather than personal opinions.

Qualities of a Good Research

- **Replicable, reproducible, and transparent:** Replicability and reproducibility are often used interchangeably. Research is reproducible if researchers achieve consistent results using the same data and analysis methods.

- **Representative and Generalizable:** Representativeness refers to a sample's ability to represent a larger group (population). By using proper methods to create a representative sample, researchers can ensure that their findings can be **generalized** to the larger population represented.

- **External validity:** There are two types of external validity: population validity and ecological validity. Research with findings that can be generalized from the sample to the larger population has a high population validity. Ecological validity can be achieved if you can apply your study's findings to real-world situations and settings.

- **Acknowledges its limitations and provides suggestions for future research:** Researchers must acknowledge their study's limitations and potential flaws and present these along with the study's findings and conclusion.

Research Challenges in Uganda

Despite its importance, research activities in Uganda face numerous challenges that limit the production, quality, and impact of research outcomes.

1. Limited Research Funding

- Research requires financial resources to support:
  - Data collection
  - Fieldwork
  - Laboratory experiments
  - Equipment
  - Publication and dissemination of findings
- Many universities and research institutions rely heavily on:
  - Government funding
  - International donors
  - Development partners
- However, government investment in research remains relatively low compared to developed countries. As a result:
  - Many research projects remain unfunded
  - Researchers struggle to conduct large-scale studies
  - Some research initiatives depend on external donor priorities rather than local needs

Research Challenges in Uganda

2. Inadequate Research Infrastructure

- Research infrastructure includes laboratories, libraries, research centers, internet facilities, and technological equipment.
- However, in many institutions in Uganda:
  - Laboratories may lack modern equipment
  - Libraries may have limited access to current journals and books
  - Internet connectivity may be unreliable
  - Research software and tools may be unavailable
- This limits the ability of researchers and students to conduct high-quality and competitive research.

3. Limited Access to Research Data

- Reliable and accurate data is essential for effective research. However, in Uganda researchers often face challenges such as:
  - Lack of updated national statistics
  - Limited access to institutional databases
  - Poor record keeping in some organizations
  - Restrictions in accessing government data

Research Challenges in Uganda

4. Low Research Culture in Some Institutions

- Research culture refers to the environment that encourages research activities among academics, students, and institutions. However, in some institutions:
  - Teaching responsibilities are very heavy
  - Limited incentives are provided for research
  - Few research collaborations exist
- As a result:
  - Some academics focus mainly on teaching rather than research
  - Students may graduate with limited research exposure
  - Research output remains low.

Research Challenges in Uganda

5. Brain Drain

- Brain drain refers to the migration of skilled researchers and academics to other countries. Many talented Ugandan researchers move abroad because of:
  - Better salaries
  - Advanced research facilities
  - More funding opportunities
  - Greater professional recognition
- This leads to:
  - Loss of skilled human resources
  - Reduced mentorship for young researchers
  - Slower development of local research capacity.

6. Limited Collaboration Between Universities and Industry

- Effective research often requires collaboration between, universities, government institutions, private sector organizations, and international research bodies
- In Uganda, such collaborations are still limited and these lead to:
  - Research findings not being applied in industry
  - Limited commercialization of research innovations
  - Weak knowledge transfer between academia and industry.

Research Challenges in Uganda

7. Bureaucratic and Administrative Barriers

- Researchers sometimes face administrative challenges such as:
  - Long approval processes for research permits
  - Delays in ethical clearance
  - Institutional bureaucracy
- These processes may delay research projects and discourage researchers from pursuing certain studies.

8. Limited Research Skills Among Some Students

- Many students, especially at undergraduate level, face challenges in conducting research due to:
  - Limited training in research methods
  - Weak statistical and analytical skills
  - Limited experience in academic writing
  - Poor understanding of data analysis tools
- This affects the quality of student research projects and dissertations.

Research Challenges in Uganda

9. Difficulty in Publishing Research

- Publishing research findings in reputable journals can be difficult due to:
  - High publication fees
  - Strict peer review standards
  - Limited experience in academic writing
  - Lack of mentorship in scholarly publishing
- As a result, many researchers struggle to disseminate their findings globally.

10. Limited Use of Technology in Research

- Although digital technologies have improved research globally, some challenges remain in Uganda such as:
  - Limited access to advanced research software
  - Lack of training in data science and artificial intelligence tools
  - Limited digital research databases
- This restricts the adoption of modern research methods.

Research Challenges in Uganda

11. Ethical and Integrity Issues

- Some research challenges relate to ethics, including:
  - Plagiarism
  - Data fabrication
  - Poor citation practices
  - Lack of adherence to ethical research standards
- These issues can damage the credibility of research institutions and reduce trust in research findings.

Qualities of a Good Researcher

- **Curiosity:** A good researcher
  - Asks questions constantly
  - Identifies gaps in knowledge
  - Seeks deeper explanations
  - **Note:** Curiosity is the foundation of research. Without it, no meaningful research question can emerge.

- **Critical thinking:** This thinking involves
  - Questioning assumptions
  - Evaluating evidence logically
  - Distinguishing facts from opinions
  - Identifying weaknesses in arguments
  - **Note:** A researcher must not accept information blindly.

- **Analytical Skills:** A good researcher
  - Interprets data correctly
  - Understands research design
  - Applies appropriate statistical tools
  - Draws logical conclusions
  - **Note:** Poor analysis leads to misleading findings.

- **Creativity and Innovation:** Research requires
  - Generating original ideas
  - Designing new methods
  - Solving problems creatively
  - **Note:** Innovation makes research impactful.

Qualities of a Good Researcher

- **Patience**
  - Research is time-consuming:
  - Data collection may take months
  - Experiments may fail
  - Publication may require revisions
  - **Note:** Patience is essential for long-term success.

- **Perseverance:** A good researcher
  - Does not give up easily
  - Learns from rejection
  - Improves work after criticism
  - Rejections are part of academic growth.

- **Attention to Detail:** Research demands accuracy in
  - Data recording
  - Referencing
  - Methodological steps
  - Statistical calculations
  - **Note:** Small errors can invalidate entire studies.

- **Integrity:** A good researcher
  - Reports findings honestly
  - Avoids data fabrication
  - Avoids falsification
  - Integrity builds credibility.

Qualities of a Good Researcher

- **Respect for Participants:** Ethical research requires
  - Informed consent
  - Confidentiality
  - Protection of participants
  - **Note:** Ethics approval is often mandatory in academic research.

- **Avoidance of Plagiarism:** Researchers must:
  - Cite sources properly
  - Acknowledge other scholars
  - Present original work
  - Plagiarism damages academic reputation.

- **Good Communication Skills:** A researcher should
  - Write clearly and concisely
  - Present findings effectively
  - Defend research logically
  - **Note:** Research is incomplete if it cannot be communicated.

- **Organizational Skills:** A good researcher:
  - Plans research stages
  - Manages time effectively
  - Keeps systematic records
  - Meets deadlines
  - **Note:** Good organization increases efficiency.

Qualities of a Good Researcher

- **Continuous Learning:** Research fields evolve. A researcher must:
  - Read current literature
  - Learn new tools and technologies
  - Attend conferences and seminars
  - **Note:** Learning never stops in academia.